R0004E Exercise 4 2025

Final answers

1. –

2. –

3. $G_p(s)H(s) = \dfrac{4}{s(s+1)(s+2)}$, $H(s) = 1$

Specs : ϕ_M (PM) $= 50^\circ$ Define $G_p \cdot H = G_T$

$T_s < 4$ sec

From Bode plot, $GM \approx 3\,dB$ at 1.5 rad/s

$\phi_M = PM \approx 10^\circ$ at 1.3 rad/s

To get PM = 50, need

$\arg G_pH = \arg G_T < -180 + \phi_M = -180 + 50 = -130$ at ω_1

using formula (this formula is used if the specs require certain T_s)

$\omega_1 = \dfrac{8}{T_s \tan\phi_m} = \dfrac{8}{4\tan(50^\circ)} = \dfrac{2}{\tan(50)} = \dfrac{2}{1.1918} \approx 1.6782 \approx 1.68 \approx 1.7$

at ω_1

$|G_T(j\omega_1)| \approx 0.5$, $\angle G_T(j\omega_1) \approx -190^\circ$

To achieve ϕ_M of 50° at ω_1, we need to increase the phase as much as θ

$\theta = \arg G_c(j\omega_1) = -180 + \phi_M - \arg G_T(j\omega)$

$= -180 + 50 + 190 = 60^\circ$

$\phi_{max} = \theta = \sin^{-1}\dfrac{1-\alpha}{1+\alpha}$ $\quad 0.866 = \dfrac{1-\alpha}{1+\alpha}$

$\sin\theta = \dfrac{1-\alpha}{1+\alpha}$ $\quad 0.866 - 1 = -0.866\alpha - \alpha \Rightarrow \alpha = 0.0718 \approx 0.072$

The resulting magnitude :

$$|U(j\omega)|_{dB} = \frac{1}{\sqrt{\alpha}} = \frac{1}{\sqrt{0.072}} = 3.7268 \text{ dB}$$

which is equivalent to

$$|U(j\omega)| = 10^{(3.7268/20)} = 1.5358$$

From Bode plot, at magnitude 3.7268 dB the freq :

$\omega_{max} \approx 1.2$ rad/s

$$\omega_{max} = \frac{1}{T\sqrt{\alpha}} \Rightarrow T = \frac{1}{\sqrt{\alpha}} \cdot \frac{1}{\omega_{max}} = \frac{1.5358}{1.2} \approx 1.2798 \approx 1.28$$

Hence, Break frequency at

$$\frac{1}{T} = \frac{1}{1.28} \approx 0.7813 \quad \text{and} \quad \frac{1}{\alpha T} = \frac{1}{1.28 \times 0.072} = 10.852$$

Hence, the lag compensator is

$$U(s) = \frac{1}{\alpha} \frac{s + \frac{1}{T}}{s + \frac{1}{\alpha T}} = \frac{1}{0.072} \cdot \frac{s + 0.7813}{s + 10.852} =$$

$$= 13.9 \frac{s + 0.7813}{s + 10.852}$$

$$U(s) \approx \frac{14s + 10.852}{s + 10.852}$$

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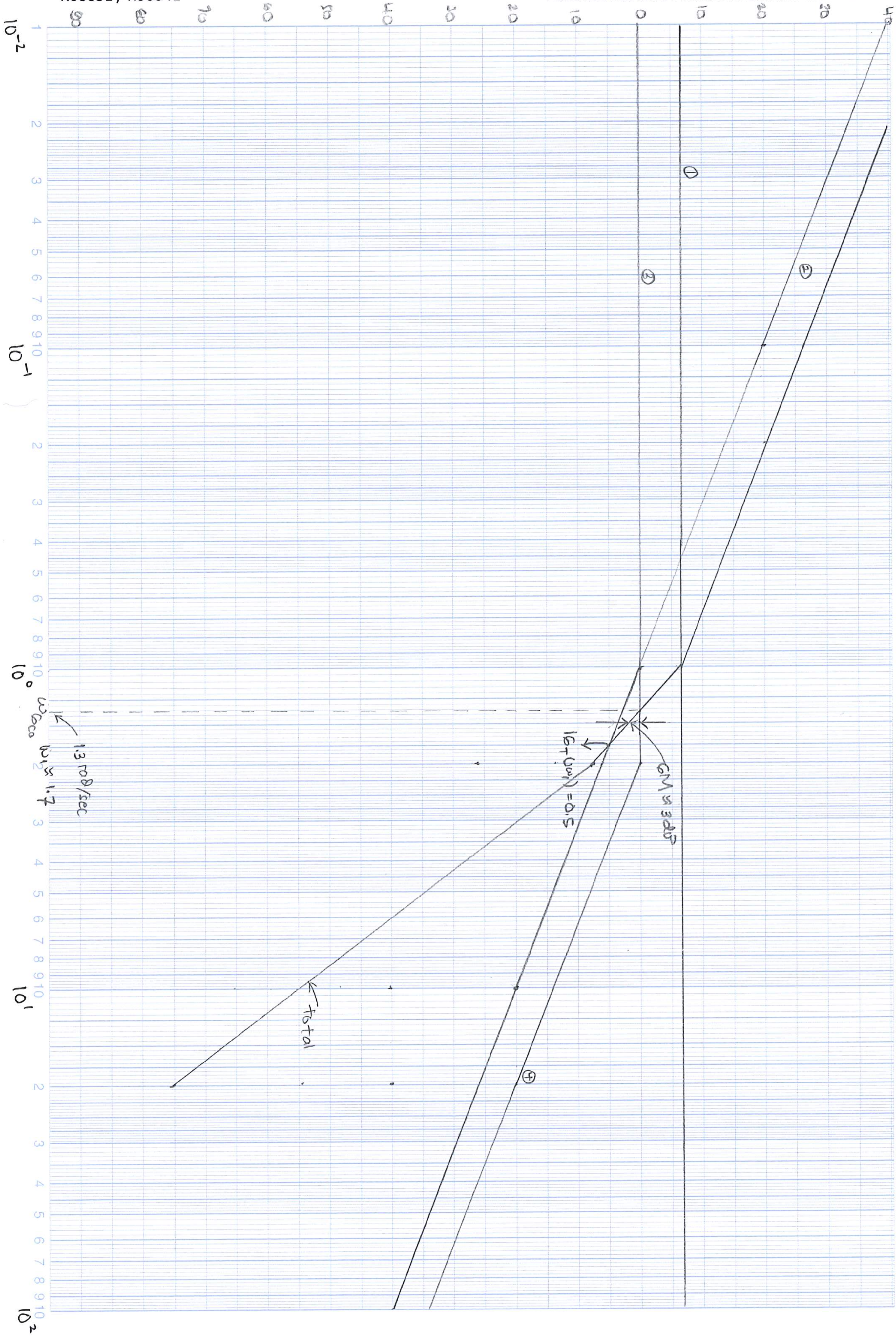

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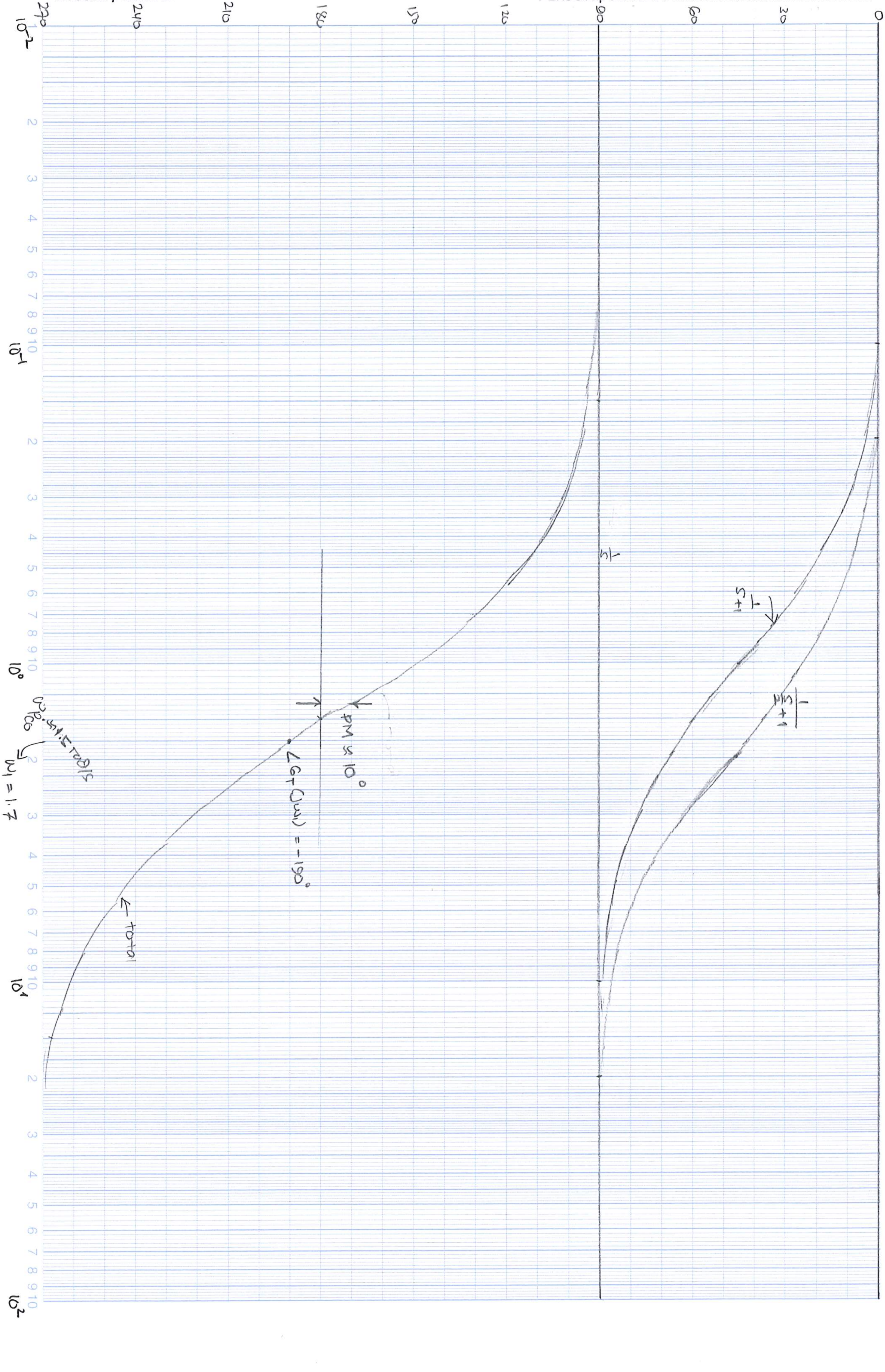

4 $G(s) = \dfrac{K_g}{(s+2)(s+\alpha)}$

a. CLTF with unity FB

$$T(s) = \frac{K_g}{(s+2)(s+\alpha)+K_g}$$

b. $\alpha > 13 \quad \vee \quad K_g < \frac{2}{9}\alpha$

c To improve damping a PD controller needs to be used.

5 $G(s) = \dfrac{50}{s(s+3)(s+6)}$

See nyquist plot table column 1, row 5

crossing with real axis at ~~(-0.3, j0)~~ (-0.3, j0) or $(-\frac{25}{81}, j0)$

$P = 0$, $N = 0$ $\quad Z = P + N = 0 \Rightarrow$ stable

6 $G(s) = \dfrac{-(s^2+4s+6)}{s^2+5s+4}$

$P = 0$

$N = 1$

$Z = P + N = 1 \Leftarrow$ unstable

7. $G(s) = \dfrac{K}{(s+2)(s+4)(s+6)}$ $\quad K = 1$ $\quad$ c. $0 < K \leq 480$

a. Crossing real axis at $(0.0208, j0)$, $(-0.0021, j0)$

$\omega = 0$ $\quad \omega = \sqrt{44}$

b. Combine (a) and crossing ~~at~~ imaginary axis at $(0, \pm 0.012)$

~~$\omega = 2$~~

See nyquist plot table column 2, row 4